Bacterial Infections

36

Dina Averbuch

36.1 Introduction (Tomblyn et al. 2009; Averbuch et al. 2017, 2020; Misch and Andes 2019; Dandoy et al. 2020; Martinez-Nadal et al. 2020; Vu et al. 2020)

The most frequent bacterial infections in HCT patients are bloodstream infections (BSI). They occur in 5–10% of auto-HCT and 20–50% of allo-HCT patients, with higher rates before engraftment, and are associated with increased morbidity and mortality. BSI can be divided into primary and secondary. Primary BSIs are mainly central line catheter-related (CRBSI) or mucosal barrier injury-associated (MBI-BSI). The latter occurs when the pathogen originates from the oral cavity or GI (e.g., Enterobacterales) in patients with neutropenia and/or MBI (GI GVHD grade 3/4, severe diarrhea). Secondary BSI accompanies site-specific infection (e.g., *Pseudomonas aeruginosa* pneumonia and BSI). Other frequently clinically documented bacterial infections include pneumonia, skin and soft tissue infection, and typhlitis. Patient-related risk factors for bacterial infections include older age, comorbidities, low functional capacity, high-risk hematological disease (active malignancy, aplastic anemia in allo-HCT, lymphoma, or leukemia in auto-HCT), intestinal domination of specific bacteria, and decreased microbiome diversity. Transplant-related risk factors are specific to the post-HCT period. During the early pre-engraftment phase, neutropenia and disruption of anatomical barriers (mucosal damage and vascular devices) predispose to infection resulting from Gram-positive cocci (GPC) and GNB. Haploidentical HCT and CBT are associated with slower engraftment, delayed immune reconstitution, and higher infection risk. During the intermediate phase, starting at engraftment (+30 to +100 days), the main risk factors are CVC, lack of immune reconstitution, and GVHD-related factors, including its severity and treatment (e.g., etanercept, steroid-resistant disease). Later, humoral and cellular immunodeficiency predisposes to infections due to encapsulated pathogens (*Streptococcus pneumoniae* and *Haemophilus influenzae*), and, rarely, infections due to *Mycobacteria, Nocardia, Listeria*, and others. The main risk factors for BSI-associated mortality include older age, high comorbidity score, steroid use, persistent neutropenia, ICU-acquired BSI, shock at presentation, multi-drug resistant (MDR) bacteria (non-susceptible to ≥1 agent in ≥3 therapeutically relevant antimicrobial categories), and inadequate empirical treatment.

D. Averbuch (✉)
Faculty of Medicine, Hebrew University of Jerusalem, Jerusalem, Israel

Pediatric Infectious Diseases Unit, Hadassah University Hospital, Jerusalem, Israel
e-mail: adiana@hadassah.org.il

© The Author(s) 2024
A. Sureda et al. (eds.), *The EBMT Handbook*, https://doi.org/10.1007/978-3-031-44080-9_36

36.2 Epidemiology and Treatment of Bacterial Infections (Averbuch et al. 2017; Misch and Andes 2019; Dandoy et al. 2020; Martinez-Nadal et al. 2020; Vu et al. 2020; Trecarichi et al. 2023)

GNB has become an increasingly common cause of BSI, although sizable intercenter variability in epidemiology is reported. Attributable mortality is usually higher in patients with GNB-BSI than GPC-BSI. Increase in MDR GNB infections limits treatment options, leads to inappropriate empirical therapy, and increases mortality. The prevalence of resistance is influenced by local antibiotic use policies (prophylaxis and treatment), infection control measures, and resistance patterns throughout the specific hospital and countrywide. In the multinational prospective EBMT study, half GNBs were resistant to non-carbapenem β-lactams, the first-line treatment for febrile neutropenia; 18.5% was carbapenem-resistant (CR); and 35% was MDR. Higher resistance rates were reported in allo- versus auto-HCT patients, and in southeastern as compared with northwestern Europe. The main risk factors for carbapenem-resistant GNB are prior colonization or infection with the same bacteria, breakthrough on carbapenems, and ICU hospitalization. In this session, we address infections caused by the most frequent GPC and GNB and the main treatment options for the severe infections they cause. There are differences in recommendations and levels between American and European guidelines.

36.3 Gram-Positive Infections

Coagulase-negative Staphylococcus (CoNS) causes 7–63% of BSI. True CoNS BSI, defined as at least two consecutive positive blood cultures, is usually CVC-related. Methicillin resistance is frequent, prompting treatment with glycopeptides. The prognosis is usually good.

Staphylococcus aureus causes 3–8% of BSI, and its attribute mortality is high (12–40%). Cefazolin and oxacillin are the therapeutic mainstays against methicillin-susceptible *Staphylococci*. Methicillin-resistant *Staphylococcus aureus* (MRSA) is frequent in some countries but not all. The main treatment options for MRSA BSI include vancomycin and daptomycin, with a higher daptomycin dose (8–12 mg/kg/day) considered for BSI treatment. As daptomycin is inactivated by surfactant, it should not be used to treat pneumonia. Reduced vancomycin susceptibility has been reported (VISA), with some strains daptomycin nonsusceptible. Other active agents for specific indications (pneumonia and/or skin and soft tissue infections) include ceftaroline, lipoglycopeptides, delafloxacin, and linezolid (linezolid is myelosuppressive).

Enterococci cause 4–24% of BSI, usually occurring later after HCT. *E. faecium* is more common than *E. faecalis*. Risk factors for the vancomycin-resistant Enterococci (VRE) BSI include previous colonization, mucositis, and broad-spectrum antimicrobial exposure. It typically occurs in patients in poor clinical condition and in association with GVHD, perhaps explaining its high associated mortality and the failure of VRE-active empirical therapy to reduce that mortality. The main treatment option for ampicillin-susceptible enterococcal infection is ampicillin; for ampicillin-resistant infection, vancomycin; and for VRE, linezolid or daptomycin or a combination of daptomycin with either ampicillin, ceftriaxone, or ceftaroline. Reduced daptomycin susceptibility has been reported among VRE; increased dosage is thus recommended.

Streptococcus viridans (VS) causes 7–24% of BSI, usually occurring soon after HCT. VS BSI can be accompanied by ARDS and septic shock associated with high mortality. Mucositis, especially following cytarabine, exposure to fluoroquinolones (FQ) or ceftazidime, antiacids, MAC, and haploidentical HCT, predispose to VS BSI. VS is susceptible to most β-lactams used empirically for febrile neutropenia except for ceftazidime. The possibility of β-lactam-resistant VS infections, mainly observed after exposure to β-lactams or in nosocomial BSI, justifies the vancomycin addition in neutropenic patients with septic shock.

HCT patients are at risk for *invasive pneumococcal disease* (IPD). The main presentations are BSI, pneumonia, and meningitis, occurring late (sometimes years) after HCT, with a mortality of 13–30%. Predisposing factors include allo- versus auto-HCT, hypogammaglobulinemia, and cGVHD. Vaccination against IPD is important. Antibiotic prophylaxis, preferably with oral penicillin (if local penicillin resistance rates are low), shall be considered in patients with cGVHD on immune suppressive therapy and those with hypogammaglobulinemia who do not receive replacement therapy, regardless of prior vaccinations.

36.4 Gram-Negative Infections (Paul et al. 2022; Tamma et al. 2022)

GNB infecting HCT patients include mainly *Enterobacterales* (~70%) and non-fermentative GNB (NFGNB, ~24%). *E. coli* is the most frequent GNB prior to engraftment; after engraftment, increase is seen in the proportion of non-*E. coli* Enterobacteriaceae (with higher resistance rates) and NFGNB. GNB infections may present with BSI, sepsis/septic shock, pneumonia, enterocolitis, and soft tissue infections (e.g., ecthyma gangrenosum, typically *Pseudomonas aeruginosa*-associated). Without appropriate supportive and antibiotic treatment, death may occur within hours. The main GNB, their resistance pattern, and treatment recommendations are addressed below.

36.4.1 Broad-Spectrum β-Lactamase-Producing *Enterobacterales*

The main resistance mechanism to empirical therapy in *Enterobacterales* is broad-spectrum β-lactamase production, due to:

1. Extended-spectrum β-lactamase production (ESBL-E; 2–44% of *Enterobacterales* in HCT patients). They are typically resistant to ceftriaxone, but in vitro can appear susceptible to other non-carbapenem β-lactams. The treatment of choice is carbapenem. Piperacillin-tazobactam, ceftazidime, or cefepime are not recommended for treating severe ESBL-E infections, even if susceptibility is demonstrated.
2. AmpC production, that can *be inducible* (*Enterobacter cloacae, Klebsiella aerogenes, Citrobacter freundii), meaning resistance can develop on treatment with* non-carbapenem β-lactam. Carbapenem is a treatment of choice. Cefepime can be used if the bacteria are fully susceptible and ESBL production is excluded.

European guidelines recommend carbapenem as a treatment of choice for patients with severe infection resulting from third-generation cephalosporines-resistant Enterobacterales.

36.4.2 Carbapenemase-Producing Enterobacterales (CPE)

The main carbapenem resistance mechanism in *Enterobacterales* is carbapenemases production, including *Klebsiella pneumoniae* carbapenemase (KPC), metallo-β-lactamases (New Delhi metallo-β-lactamase (NDM), VIM, etc.), and OXA-48-like enzymes. Among Enterobacterales, carbapenem resistance is more frequent in *Klebsiella pneumoniae*. Several new β-lactams are active against carbapenemase-producing Enterobacterales; clinical data on treatment with some of them and specifically data in immunocompromised patients with BSI is, however, very limited, and resistance has been reported on treatment or regardless of previous exposure.

Severe infections due to KPC-producing Enterobacterales can be treated with meropenem-vaborbactam or ceftazidime-avibactam. Other options are imipenem-relebactam and cefiderocol, but clinical data on their use is very limited. OXA-48-like-producing *Enterobacterales* can be treated with ceftazidime-avibactam or alternatively with cefiderocol. Metallo-β-lactamases-producing Enterobacterales can be treated with aztreonam plus a ceftazidime-avibactam combination, or

with cefiderocol. Treatment with meropenem-vaborbactam or ceftazidime-avibactam has been associated with better outcomes than the older (mainly colistin-based) regimens. Polymyxins are not suggested for the treatment of CRE infections susceptible to β-lactams. β-lactams nephrotoxicity rates were significantly lower than that of polymyxin-based or aminoglycoside-based regimens. A routine combination of β-lactams with aminoglycosides, polymyxins, or FQ is not recommended. For patients with severe infections caused by CRE, susceptible in vitro only to polymyxins, aminoglycosides, tigecycline, or fosfomycin, or in the case of non-availability of new antibiotics, treatment with more than one active drug is suggested.

36.4.3 *Pseudomonas aeruginosa* (PA)

PA causes 4–16% of BSI with a 39–79% mortality, especially in ICU-acquired and resistant PA infections. PA in HCT patients is frequently resistant to β-lactams, FQ, and aminoglycosides; 25–71% are MDR, and resistance can develop on treatment. "Difficult-to-treat" resistant (DTR) PA is not susceptible to any of the following: piperacillin-tazobactam, ceftazidime, cefepime, aztreonam, meropenem, imipenem-cilastatin, ciprofloxacin, and levofloxacin. Non-carbapenem β-lactams treatment (i.e., piperacillin-tazobactam, ceftazidime, cefepime, and aztreonam) is preferred for infections caused by susceptible PA. To treat carbapenem-resistant but non-carbapenem β-lactam-susceptible PA infection, high-dose extended infusion of non-carbapenem β-lactam can be used. Ceftolozane-tazobactam treatment is preferred for patients with DTR PA, as outcomes are better and nephrotoxicity rates lower than on other regimens (mainly polymyxin- and aminoglycoside-based), according to observational studies. Ceftolozane-tazobactam resistance, with ceftazidime-avibactam cross-resistance, can develop on treatment. Other treatment options for DTR PA include imipenem-relebactam and ceftazidime-avibactam. An alternative is cefiderocol, but clinical data on its use are limited. β-lactam combination with aminoglycosides, FQ, or polymyxins is not routinely recommended for β-lactam-susceptible PA infections. Severe PA infections, especially in neutropenic patients, are, however, frequently treated with combination therapy, at least until the patient stabilizes, as prognosis is poor. Nephrotoxicity remains a concern. For DTR PA infections resistant to all β-lactams, a combination of less in-vitro resistant β-lactam with an aminoglycoside or polymyxin B can be considered. If severe CRPA infections are treated with polymyxins, aminoglycosides, or fosfomycin, two in-vitro active drugs should be considered.

36.4.4 Other NFGNBs

Other NFGNBs are rarely responsible for infection in HCT patients. *Stenotrophomonas maltophilia* can cause severe hemorrhagic pneumonia or BSI. It is resistant to multiple antibiotics, with intrinsic resistance to carbapenems and aminoglycosides. TMP/SMX is a treatment of choice, but resistance has been reported, and the sulfonamide can be poorly tolerated. For severe infections, a combination with one of minocycline (preferred)/tigecycline/levofloxacin/cefiderocol is an option. Theoretically, a combination of ceftazidime-avibactam with aztreonam can overcome resistance mechanisms.

Infections resulting from carbapenem-resistant (CRAB) and MDR *Acinetobacter baumannii* are typically ICU-associated, with mortality in HCT patients of 49–95%. Recommended therapy in patients with severe CRAB infections is a combination of at least two in-vitro active agents. High-dose extended/continuous ampicillin-sulbactam infusion is a preferred therapy, to be considered even when susceptibility is not demonstrated. Agents generally used in combination include minocycline (preferred), tigecycline (alternative, high dose), and polymyxins. Some consider the addition of high-dose, extended-infusion meropenem as a part of a combination for infections caused by non-highly resistant CRAB, with a step-down to single agent after improvement. Cefiderocol treatment has been associated with higher mortality than other therapies and should not be used routinely. Sulbactam-durlobactam is a novel combination with promising activity against CRAB.

36.5 Bacterial Infection Syndromes (Freifeld et al. 2011; Misch and Andes 2019; McDonald et al. 2018; Schmidt-Hieber et al. 2018; van de Beek et al. 2016)

36.5.1 Central Line-Related BSI (CRBSI)

CRBSI should be suspected when blood cultures are persistently positive, at the presence of exit site or tunnel infection, and when fever and chills develop during CVC flushing. This can be proved by a differential time to positivity of >120 min in blood cultures simultaneously drawn from the CVC and a vein; or positive semiquantitative or quantitative cultures. Catheter removal, in addition to systemic antimicrobial therapy, is recommended. CVC salvage in stable patients without complications can be attempted by an antimicrobial lock.

36.5.2 Pneumonia

Bacterial pneumonia occurs in 11–24% of patients, its specific etiology frequently unidentified. During neutropenia, it results from GNB (including PA) and GPC typical to this stage. Specific entities include *Stenotrophomonas maltophilia* hemorrhagic pneumonia, VS-associated ARDS, and nosocomial legionellosis. In the late post-engraftment phase, IPD and *Haemophilus influenzae* should be considered. With symptoms and signs often atypical and scarce, it can progress rapidly. Hypoxemia can be the sole finding and should prompt chest CT, diagnostic bronchoscopy if feasible, and immediate empirical antibiotic therapy, reflecting the colonization history and local resistance patterns. Nebulized antibiotic use for the DTR-*P. aeruginosa* pneumonia treatment is not routinely recommended.

36.5.3 Diarrhea

Clostridioides difficile-associated infection (CDI) occurs in 5–30% of HCT patients following exposure to broad-spectrum antibiotics and chemotherapy. Clinical manifestation may be paradoxically mild, but severe complications, such as toxic megacolon and perforation, can occur. Diagnosis is usually based on a combination of a glutamate dehydrogenase enzyme immunoassay (EIA), a nucleic acid amplification test (NAAT), and a toxin A and B EIA. Treatment choice is determined by the CDI severity, the patient's ability to take oral treatment and whether it is a recurrent infection.

Bacterial diarrhea due to *Shigella*, *Salmonella*, *Yersinia*, *Campylobacter* spp., and enterohemorrhagic *E. coli* is rare, usually occurring in a community-acquired setting. Routine stool culture is thus recommended for patients with diarrhea only within 3 days of admission.

Hygiene measures and contact precautions are important to prevent intestinal pathogen spread.

36.5.4 CNS Infections

Bacteria rarely cause brain abscesses (*Streptococcus viridans*, *Staphylococcus aureus*, *Klebsiella pneumoniae*) or meningoencephalitis (*Listeria*, IPD). Clinical manifestations include fever, headache, altered mental state, and focal neurological signs and seizures. These can be subtle due to decreased inflammation. MRI is more sensitive than CT in identifying CNS infection. When bacterial meningitis is suspected, empirical therapy should include cefotaxime/ceftriaxone plus amoxicillin or ampicillin or penicillin G. Vancomycin addition can be considered based on local rates of *Streptococcus pneumoniae* ceftriaxone resistance.

36.6 General Principles of Management of Bacterial Infections (Freifeld et al. 2011; Averbuch et al. 2013)

1. Empirical antibiotic therapy should be started immediately when bacterial infection is suspected. It should reflect the patient's clinical condition, prior colonization or infection with resistant bacteria, and local epidemiology.

Monitoring local bacterial resistance patterns and patient colonization status in endemic settings is important.

2. A broad-spectrum antibiotic regimen (e.g., meropenem or β-lactam with aminoglycoside with/without vancomycin) should be used if the patient is unstable or previously colonized/infected with resistant bacteria. This should be followed by de-escalation to narrower spectrum monotherapy in stable patients once a resistant infection is ruled out. Novel laboratory techniques (e.g., matrix-assisted laser desorption/ionization-time of flight, MALDI-TOF) speed pathogen identification and antibiotic susceptibility testing.
3. Targeted therapy should be based on the susceptibility profile using the narrowest spectrum and least toxic active antibiotic. Consultation with a specialist in infectious diseases is recommended, especially when treating infections due to resistant bacteria.
4. *Antimicrobial stewardship* is important in limiting unnecessary antibiotic exposure and in optimizing antimicrobial therapy based on pharmacokinetic/pharmacodynamic principles with TDM, whenever available. A loading dose, followed by prolonged or continuous infusion of time-dependent antibiotics (e.g., β-lactams) has been associated with lower mortality than short-term infusion, along with lower rates of resistance development on treatment.
5. Source control is important. CVC removal is recommended in:
   (a) Infections due to *S. aureus, P. aeruginosa,* fungi, or mycobacteria
   (b) Severe sepsis with hemodynamic instability
   (c) Suppurative thrombophlebitis
   (d) Endocarditis
   (e) Persistently positive (>72 h) blood cultures under appropriate antibiotics
   (f) Tunnel infection or port pocket site infection
6. Antibiotic treatment should be continued for at least 7 days until the infection is microbiologically eradicated and all clinical signs resolved, with the patient afebrile for at least 4 days. Several studies and meta-analyses have failed to demonstrate difference in mortality in patients who received short (7–10 days) vs. long (<10 days) antibiotic treatment for GNB BSI.

36.7 Prevention of Bacterial Infections (Tomblyn et al. 2009; Mikulska et al. 2018; Egan et al. 2019; Buetti et al. 2022)

General infection prevention measures include personal patient hygiene, bathing with chlorhexidine-impregnated washcloths, and use of single-patient rooms. Important infection control measures include standard precautions, especially hand hygiene, use of gloves and gowns when soiling is likely, and environmental cleaning. Multifaceted interventions should be practiced preventing MDR bacteria spread, including patient screening for colonization in the epidemic setting, using contact precautions, isolation, and cohorting of colonized and/or infected patients and staff (this last, for CPE-colonized patients). Routine CPE-targeted decolonization with non-absorbable oral antibiotics is not supported.

The main elements of *CRBSI prevention bundles* include sterile insertion by a specialized team, avoidance of femoral sites, chlorhexidine cleaning during use, antiseptic- or antimicrobial-impregnated CVC, and removal of unnecessary catheters.

FQ prophylaxis (FQP) is currently recommended by several national and international guidelines in adult patients with expected neutropenia ≥7 days, aiming to reduce all-cause mortality, febrile episodes, and GNB-BSI rates. An increase in FQ-resistant GNB affects FQP efficacy, however. Meta-analysis of studies published 1980–2018 shows no reduction in mortality on FQP. Febrile neutropenia rates fell by ~15% on FQP, and reduction in BSI rates was observed mainly in acute leukemia and auto-HCT patients, but not in allo-HCT patients. Possible FQP benefits should be weighed against its potential harm, including CDI, microbiome alterations predisposing to increased GVHD risk,

side effects, and association with colonization/infection with FQ-resistant or MDR GNB.

Late infection prevention (>100 days post-HCT), targeting mainly encapsulated bacteria, includes:

1. Oral prophylaxis with penicillin (or other agents, according to local antibiotic resistance patterns) in patients with cGVHD or hypogammaglobulinemia.
2. IVIg in patients with severe hypogammaglobulinemia (serum IgG level < 400 mg/dL);
3. Vaccination.

Key Points

- An increase in infections due to resistant GNB, such as ESBL Enterobacterales, carbapenemase-producing Enterobacterales (CPE), MDR GNB, or DTR *Pseudomonas aeruginosa,* leads to delay in appropriate therapy and increases mortality.
- Main targeted therapy options for severe infections caused by resistant GNB include:
  - ESBL Enterobacterales: carbapenems.
  - KPC-producing Enterobacterales*:* meropenem-vaborbactam or ceftazidime-avibactam.
  - OXA-48-like-producing Enterobacterales: ceftazidime-avibactam.
  - Metallo-β-lactamases-producing Enterobacterales: aztreonam plus ceftazidime–avibactam, or cefiderocol.
  - DTR *Pseudomonas aeruginosa*: ceftolozane-tazobactam.
  - Routine combination therapy of β-lactams with aminoglycosides/FQ/polymyxins for infection due to MDR GNB susceptible to β-lactam is not recommended (with a possible exception of a severe infections due to *Pseudomonas aeruginosa in neutropenic patients*).
  - High-dose prolonged β-lactam infusion can maximize efficacy.
- Antimicrobial stewardship aims to individualize an empirical approach to patients with suspected infection (escalation vs. de-escalation), limiting unnecessary antibiotic use, and optimizing treatment based on pharmacokinetic/pharmacodynamic principles.
- Infection control is crucial to limit the spread of MDR pathogens.
- Fluoroquinolone prophylaxis is controversial.
- Encapsulated bacteria (*Streptococcus pneumoniae* and *Haemophilus influenzae*) cause infection during the late post-engraftment period. Preventive measures include oral prophylaxis, IVIg, and vaccinations.

Acknowledgement J. Strahilevitz for critical revision.

References

Averbuch D, Orasch C, Cordonnier C, Livermore DM, Mikulska M, Viscoli C, Gyssens IC, Kern WV, Klyasova G, Marchetti O, Engelhard D, Akova M, et al. European guidelines for empirical antibacterial therapy for febrile neutropenic patients in the era of growing resistance: summary of the 2011 4th European conference on infections in leukemia. Haematologica. 2013;98(12):1826–35.

Averbuch D, Tridello G, Hoek J, Mikulska M, Akan H, Segundo LYS, Pabst T, Ozcelik T, Klyasova G, Donnini I, Wu D, Gulbas Z, Zuckerman T, Botelho de Sousa A, Beguin Y, Xhaard A, Bachy E, Ljungman P, de la Camara R, Rascon J, Ruiz Camps I, Vitek A, Patriarca F, Cudillo L, Vrhovac R, Shaw PJ, Wolfs T, O'Brien T, Avni B, Silling G, Al Sabty F, Graphakos S, Sankelo M, Sengeloev H, Pillai S, Matthes S, Melanthiou F, Iacobelli S, Styczynski J, Engelhard D, Cesaro S. Antimicrobial resistance in gram-negative rods causing bacteremia in hematopoietic stem cell transplant recipients: intercontinental prospective study of the infectious diseases working party of the European Bone Marrow Transplantation Group. Clin Infect Dis. 2017;65(11):1819–28.

Averbuch D, Tridello G, Hoek J, Mikulska M, Pabst T, Segundo LYS, Akan H, Ozcelik T, Donnini I, Klyasova G, Botelho de Sousa A, Zuckerman T, Tecchio C, de la Camara R, Aki SZ, Ljungman P, Gulbas Z, Nicolas-Virelizier E, Calore E, Perruccio K, Ram R, Annaloro C, Martino R, Avni B, Shaw PJ, Jungova A,

Codeluppi K, O'Brien T, Waszczuk-Gajda A, Batlle M, Pouli A, Lueck C, Gil L, Iacobelli S, Styczynski J, Engelhard D, Cesaro S. Intercontinental study on pre-engraftment and post-engraftment gram-negative rods bacteremia in hematopoietic stem cell transplantation patients: risk factors and association with mortality. J Infect. 2020;81(6):882–94.

Buetti N, Marschall J, Drees M, Fakih MG, Hadaway L, Maragakis LL, Monsees E, Novosad S, O'Grady NP, Rupp ME, Wolf J, Yokoe D, Mermel LA. Strategies to prevent central line-associated bloodstream infections in acute-care hospitals: 2022 update. Infect Control Hosp Epidemiol. 2022;43(5):553–69.

Dandoy CE, Kim S, Chen M, Ahn KW, Ardura MI, Brown V, Chhabra S, Diaz MA, Dvorak C, Farhadfar N, Flagg A, Ganguly S, Hale GA, Hashmi SK, Hematti P, Martino R, Nishihori T, Nusrat R, Olsson RF, Rotz SJ, Sung AD, Perales MA, Lindemans CA, Komanduri KV, Riches ML. Incidence, risk factors, and outcomes of patients who develop mucosal barrier injury-laboratory confirmed bloodstream infections in the first 100 days after allogeneic hematopoietic stem cell transplant. JAMA Netw Open. 2020;3(1):e1918668.

Egan G, Robinson PD, Martinez JPD, Alexander S, Ammann RA, Dupuis LL, Fisher BT, Lehrnbecher T, Phillips B, Cabral S, Tomlinson G, Sung L. Efficacy of antibiotic prophylaxis in patients with cancer and hematopoietic stem cell transplantation recipients: a systematic review of randomized trials. Cancer Med. 2019;8(10):4536–46.

Freifeld AG, Bow EJ, Sepkowitz KA, Boeckh MJ, Ito JI, Mullen CA, Raad II, Rolston KV, Young JA, Wingard JR, A. Infectious Diseases Society of. Clinical practice guideline for the use of antimicrobial agents in neutropenic patients with cancer: 2010 update by the infectious diseases society of America. Clin Infect Dis. 2011;52(4):e56–93.

Martinez-Nadal G, Puerta-Alcalde P, Gudiol C, Cardozo C, Albasanz-Puig A, Marco F, Laporte-Amargos J, Moreno-Garcia E, Domingo-Domenech E, Chumbita M, Martinez JA, Soriano A, Carratala J, Garcia-Vidal C. Inappropriate empirical antibiotic treatment in high-risk neutropenic patients with bacteremia in the era of multidrug resistance. Clin Infect Dis. 2020;70(6):1068–74.

McDonald LC, Gerding DN, Johnson S, Bakken JS, Carroll KC, Coffin SE, Dubberke ER, Garey KW, Gould CV, Kelly C, Loo V, Shaklee Sammons J, Sandora TJ, Wilcox MH. Clinical practice guidelines for Clostridium difficile infection in adults and children: 2017 update by the Infectious Diseases Society of America (IDSA) and Society for Healthcare Epidemiology of America (SHEA). Clin Infect Dis. 2018;66(7):987–94.

Mikulska M, Averbuch D, Tissot F, Cordonnier C, Akova M, Calandra T, Ceppi M, Bruzzi P, Viscoli C, L. European Conference on Infections in. Fluoroquinolone prophylaxis in haematological cancer patients with neutropenia: ECIL critical appraisal of previous guidelines. J Infect. 2018;76(1):20–37.

Misch EA, Andes DR. Bacterial infections in the stem cell transplant recipient and hematologic malignancy patient. Infect Dis Clin N Am. 2019;33(2):399–445.

Paul M, Carrara E, Retamar P, Tangden T, Bitterman R, Bonomo RA, de Waele J, Daikos GL, Akova M, Harbarth S, Pulcini C, Garnacho-Montero J, Seme K, Tumbarello M, Lindemann PC, Gandra S, Yu Y, Bassetti M, Mouton JW, Tacconelli E, Rodriguez-Bano J. European Society of Clinical Microbiology and Infectious Diseases (ESCMID) guidelines for the treatment of infections caused by multidrug-resistant gram-negative bacilli (endorsed by European society of intensive care medicine). Clin Microbiol Infect. 2022;28(4):521–47.

Schmidt-Hieber M, Bierwirth J, Buchheidt D, Cornely OA, Hentrich M, Maschmeyer G, Schalk E, Vehreschild JJ, Vehreschild M, A. W. Group. Diagnosis and management of gastrointestinal complications in adult cancer patients: 2017 updated evidence-based guidelines of the Infectious Diseases Working Party (AGIHO) of the German Society of Hematology and Medical Oncology (DGHO). Ann Hematol. 2018;97(1):31–49.

Tamma PD, Aitken SL, Bonomo RA, Mathers AJ, van Duin D, Clancy CJ. Infectious Diseases Society of America 2022 guidance on the treatment of extended-Spectrum beta-lactamase producing Enterobacterales (ESBL-E), Carbapenem-Resistant Enterobacterales (CRE), and Pseudomonas aeruginosa with Difficult-to-Treat Resistance (DTR-P. aeruginosa). Clin Infect Dis. 2022;75(2):187–212.

Tomblyn M, Chiller T, Einsele H, Gress R, Sepkowitz K, Storek J, Wingard JR, Young JA, Boeckh MJ, et al. Guidelines for preventing infectious complications among hematopoietic cell transplantation recipients: a global perspective. Biol Blood Marrow Transplant. 2009;15(10):1143–238.

Trecarichi EM, Giuliano G, Cattaneo C, Ballanti S, Criscuolo M, Candoni A, Marchesi F, Laurino M, Dargenio M, Fanci R, Cefalo M, Delia M, Spolzino A, Maracci L, Bonuomo V, Busca A, Principe MID, Daffini R, Simonetti E, Dragonetti G, Zannier ME, Pagano L, Tumbarello M, I. Hematologic Malignancies Associated Bloodstream Infections Surveillance Registry—Sorveglianza Epidemiologica Infezioni Fungine in Emopatie Maligne Group. Bloodstream infections due to gram-negative bacteria in patients with hematologic malignancies: updated epidemiology and risk factors for multidrug-resistant strains in an Italian perspective survey. Int J Antimicrob Agents. 2023;61(6):106806.

van de Beek D, Cabellos C, Dzupova O, Esposito S, Klein M, Kloek AT, Leib SL, Mourvillier B, Ostergaard C, Pagliano P, Pfister HW, Read RC, Sipahi OR, Brouwer MC, E. S. G. f. I. o. t. Brain. ESCMID guideline: diagnosis and treatment of acute bacterial meningitis. Clin Microbiol Infect. 2016;22(Suppl 3):S37–62.

Vu DL, Dayer JA, Masouridi-Levrat S, Combescure C, Boely E, Khanna N, Mueller NJ, Kleber M, Medinger M, Halter J, Passweg J, Muller AM, Schanz U, Chalandon Y, Neofytos D, van Delden C, Kaiser L, S. Swiss Transplant Cohort. Microbiologically documented infections after adult allogeneic hematopoietic cell transplantation: a 5-year analysis within the Swiss transplant cohort study. Transpl Infect Dis. 2020;22(4):e13289.

Open Access This chapter is licensed under the terms of the Creative Commons Attribution 4.0 International License (http://creativecommons.org/licenses/by/4.0/), which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license and indicate if changes were made.

The images or other third party material in this chapter are included in the chapter's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the chapter's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder.

